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MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO  
NETWORK PROTOCOLS & ARCHITECTURE LAB  
EXPERIMENT # 12-13**

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Score: \_\_\_\_\_ Signature of the Lab Tutor: \_\_\_\_\_ Date: \_\_\_\_\_

**Introduction to Static NAT & Dynamic NAT**

| LAB<br>PERFORMANCE<br>INDICATOR | SUBJECT<br>KNOWLEDGE | DATA ANALYSIS<br>AND<br>INTERPRETATION | ABILITY TO<br>CONDUCT<br>EXPERIMENT | SCORE |
|---------------------------------|----------------------|----------------------------------------|-------------------------------------|-------|
| OBJECTIVE NO:                   |                      |                                        |                                     |       |

**Learning Objectives**

- Configure static NAT on router
- Translation of addresses between networks (NAT)

**NAT Fundamentals**

Network address translation (NAT) is a technique used for conversion between private IP addresses and public IP addresses.

NAT is used for two purposes.

- To enable private networks to communicate with the Internet.
- To enable communication between multiple private networks through the Internet.

**Static NAT**

Static NAT is based on unchanging one-to-one mapping between IP subnets. It is also known as simple NAT.

The command syntax for static NAT is:

**nat static global** **global-address** **inside** **host-address**

Run the command on the router's GE0/0/0 port to bind **global-address** (public IP address) with **host-address** (private IP address).

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**Working :**

- A device in the internal network sends data to an external network.
- The NAT router replaces the source private IP with the mapped public IP before forwarding the packet.
- When the reply packet comes back, the NAT router translates the destination public IP back to the original private IP.
- Communication happens seamlessly, but using the public IP on the internet.

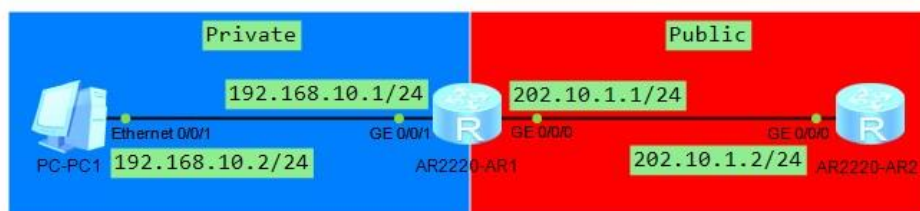
**Advantages of Static NAT:**

- Each private host always uses the same public IP.
- Useful for servers like web, mail, or FTP servers that must be reachable from outside.
- Simple to configure and understand.

**Disadvantages of Static NAT:**

- Not scalable for large networks (requires one public IP per host).
- Wastes public IP addresses.
- Manual configuration is required for each mapping.

**□ Topology**



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## Scenario

As shown in Figure, the private network is shown to the left of the router, and the public network is shown to the right of the router. To enable a PC on the private network to communicate with a server on the Internet, we need to configure **static NAT** on the router and map **192.168.0.2** (private) to **202.10.1.3** (public).

## Tasks

### Step 1 Preparing the environment

```
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname AR1
```

```
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname AR2
```

### Step 2 Configure Router AR1 and AR2

Configure the router AR1 interfaces to have IP addresses.

```
[AR1] interface GigabitEthernet 0/0/0
[AR1-GigabitEthernet0/0/0] ip address 202.10.1.1 24
[AR1-GigabitEthernet0/0/0] q
[AR1] interface GigabitEthernet 0/0/1
[AR1-GigabitEthernet0/0/1] ip address 192.168.10.1 24
[AR1-GigabitEthernet0/0/1] q
```

Configure the router AR2 interfaces to have IP addresses.

```
[AR2] interface GigabitEthernet 0/0/0
```

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```
[AR2-GigabitEthernet0/0/0] ip address 202.10.1.2 24
```

```
[AR2-GigabitEthernet0/0/0] q
```

### Step 3 Configure PCs

Configure IP address on PCs.

|              |                   |  |  |
|--------------|-------------------|--|--|
| Host Name:   | PC1               |  |  |
| MAC Address: | 54-89-98-97-71-93 |  |  |

**IPv4 Configuration**  
☒ Static ☐ DHCP ☐ Obtain DNS server address automatically  
IP Address: 192 . 168 . 10 . 2 DNS1: 0 . 0 . 0 . 0  
Subnet Mask: 255 . 255 . 255 . 0 DNS2: 0 . 0 . 0 . 0  
Gateway: 192 . 168 . 10 . 1

**IPv6 Configuration**  
☒ Static ☐ DHCPv6  
IPv6 Address: ::  
Prefix Length: 128  
IPv6 Gateway: ::

### Step 4 Verify the connectivity

```
PC> ping 202.10.1.2
```

```
Ping 202.10.1.2: 32 data bytes, Press Ctrl_C to break Request  
timeout!
```

```
Request timeout!
```

```
Request timeout!
```

```
Request timeout!
```

```
Request timeout!
```

```
--- 202.10.1.2 ping statistics ---
```

```
5 packet(s) transmitted
```

```
0 packet(s) received
```

```
100.00% packet loss
```

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## Step 5 Configure Static NAT on Router AR1

We need to configure to translate one Private IP Address to one Public IP Address.

```
[AR1] interface GigabitEthernet 0/0/0
[AR1-GigabitEthernet0/0/0] nat static global 202.10.1.3 inside 192.168.10.2
[AR1-GigabitEthernet0/0/0] q
```

Run the **display nat static** command to check whether global-address and host-address are successfully bound.

```
[AR1] display nat static
Static Nat Information:

Interface   : GigabitEthernet0/0/0
Global IP/Port : 202.10.1.3/----
Inside IP/Port  : 192.168.10.2/----
Protocol      : ----
VPN instance-name : ----
Acl number     : ----
Netmask       : 255.255.255.255
Description    : ----

Total :      1
```

From the command output, we can see that 202.10.1.3 has successfully been mapped to 192.168.0.2.

## Step 6 Verify the connectivity

```
PC> ping 202.10.1.2
```

```
Ping 202.10.1.2: 32 data bytes, Press Ctrl_C to break
From 202.10.1.2: bytes=32 seq=1 ttl=254 time=94 ms From
202.10.1.2: bytes=32 seq=2 ttl=254 time=78 ms From
202.10.1.2: bytes=32 seq=3 ttl=254 time=47 ms From
202.10.1.2: bytes=32 seq=4 ttl=254 time=31 ms From
202.10.1.2: bytes=32 seq=5 ttl=254 time=16 ms
```

```
--- 202.10.1.2 ping statistics ---
 5 packet(s) transmitted
```

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5 packet(s) received  
0.00% packet loss  
round-trip min/avg/max = 16/53/94 ms

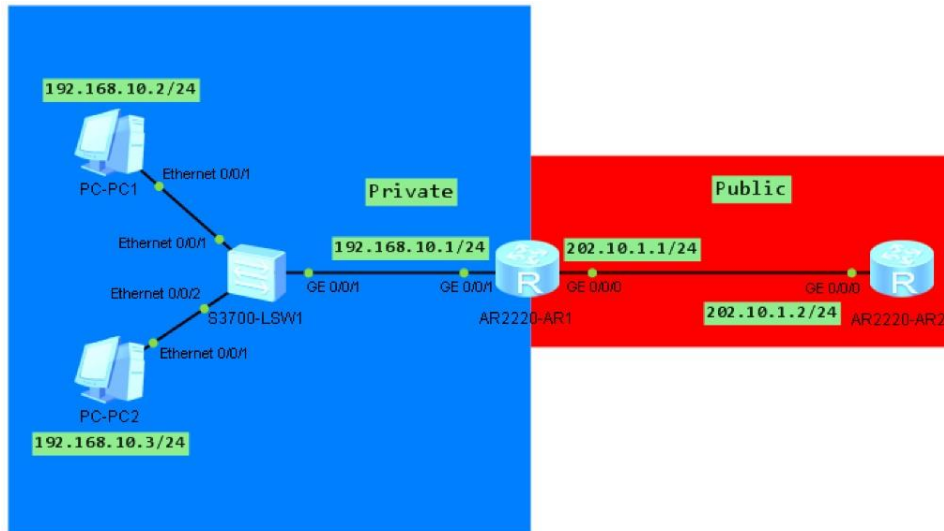
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### Dynamic NAT

The configuration of dynamic network address translations involves implementation of the nat outbound command. It relies on the prior configuration of a network access control list that is used to specify a rule to identify specific traffic to which an event or operation will be applied. An association is made between these "access control list" rules and the NAT address pool. In this manner, the addresses specified in the access control list can be translated by using the NAT address pool.

### Topology

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## Scenario

As shown in Figure, the private network is shown to the left of the router, and the public network is shown to the right of the router. To enable a PC on the private network to communicate with a server on the Internet, we need to configure **dynamic NAT** on the router and map private addresses **192.168.0.X** (private) to pool of public addresses **202.10.1.X** (public).

## Tasks

### Step 1 Preparing the environment

```
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname AR1
```

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```
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname AR2
```

## Step 2 Configure Router AR1 and AR2

Configure the router AR1 interfaces to have IP addresses.

```
[AR1] interface GigabitEthernet 0/0/0
[AR1-GigabitEthernet0/0/0] ip address 202.10.1.1 24
[AR1-GigabitEthernet0/0/0] q
[AR1] interface GigabitEthernet 0/0/1
[AR1-GigabitEthernet0/0/1] ip address 192.168.10.1 24
[AR1-GigabitEthernet0/0/1] q
```

Configure the router AR2 interfaces to have IP addresses.

```
[AR2] interface GigabitEthernet 0/0/0
[AR2-GigabitEthernet0/0/0] ip address 202.10.1.2 24
[AR2-GigabitEthernet0/0/0] q
```

## Step 3 Configure PCs

Configure IP address on PCs.

|                                                                                                                                     |                                                  |       |                                            |
|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-------|--------------------------------------------|
| Host Name:                                                                                                                          | <input type="text" value="PC1"/>                 |       |                                            |
| MAC Address:                                                                                                                        | <input type="text" value="54-89-96-97-71-93"/>   |       |                                            |
| <b>IPv4 Configuration</b>                                                                                                           |                                                  |       |                                            |
| <input checked="" type="radio"/> Static <input type="radio"/> DHCP <input type="checkbox"/> Obtain DNS server address automatically |                                                  |       |                                            |
| IP Address:                                                                                                                         | <input type="text" value="192 . 168 . 10 . 2"/>  | DNS1: | <input type="text" value="0 . 0 . 0 . 0"/> |
| Subnet Mask:                                                                                                                        | <input type="text" value="255 . 255 . 255 . 0"/> | DNS2: | <input type="text" value="0 . 0 . 0 . 0"/> |
| Gateway:                                                                                                                            | <input type="text" value="192 . 168 . 10 . 1"/>  |       |                                            |
| <b>IPv6 Configuration</b>                                                                                                           |                                                  |       |                                            |
| <input checked="" type="radio"/> Static <input type="radio"/> DHCPv6                                                                |                                                  |       |                                            |
| IPv6 Address:                                                                                                                       | <input type="text" value="::"/>                  |       |                                            |
| Prefix Length:                                                                                                                      | <input type="text" value="128"/>                 |       |                                            |
| IPv6 Gateway:                                                                                                                       | <input type="text" value="::"/>                  |       |                                            |



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|              |                                                |  |
|--------------|------------------------------------------------|--|
| Host Name:   | <input type="text" value="PC2"/>               |  |
| MAC Address: | <input type="text" value="54-89-98-3F-0D-9C"/> |  |

**IPv4 Configuration**  
☒ Static ☐ DHCP ☐ Obtain DNS server address automatically  
IP Address:  DNS1:   
Subnet Mask:  DNS2:   
Gateway:

**IPv6 Configuration**  
☒ Static ☐ DHCPv6  
IPv6 Address:   
Prefix Length:   
IPv6 Gateway:

## Step 4 Verify the connectivity

PC> `ping 202.10.1.2`

Ping 202.10.1.2: 32 data bytes, Press Ctrl\_C to break Request timeout!

Request timeout!

Request timeout!

Request timeout!

Request timeout!

--- 202.10.1.2 ping statistics ---

5 packet(s) transmitted

0 packet(s) received

100.00% packet loss

## Step 5 Configure Dynamic NAT on Router AR1

Format: `nat address-group group-index start-address end-address`

### Usage Scenario

The address pool is a **set of consecutive IP addresses**. When a packet from the private network reaches the public network through address translation, the source address of the packet will be translated to another address by the address pool.

### Precautions

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The start IP address of the address pool must be smaller than or equal to the end IP address of the address pool and up to 255 IP addresses can be configured in the address pool.

```
[AR1] nat address-group 1 202.10.1.3 202.10.1.10
[AR1] acl 2000
[AR1-acl-basic-2000] rule permit source 192.168.10.0 0.0.0.255
[AR1-acl-basic-2000] q
[AR1] interface GigabitEthernet 0/0/0
[AR1-GigabitEthernet0/0/0] nat outbound 2000 address-group 1 no-pat [AR1-
GigabitEthernet0/0/0] q
```

```
[AR1] display nat address-group 1 NAT
```

Address-Group Information:

```
-----
Index   Start-address   End-address   -----
1       202.10.1.3      202.10.1.10  -----
Total : 1
```

```
[AR1] display nat outbound NAT
```

Outbound Information:

```
-----
Interface           Acl   Address-group/IP/Interface   Type
-----
GigabitEthernet0/0/0 2000                               1   no-pat  --
-----
Total : 1
```

## Step 6 Verify the connectivity

```
PC> ping 202.10.1.2
```

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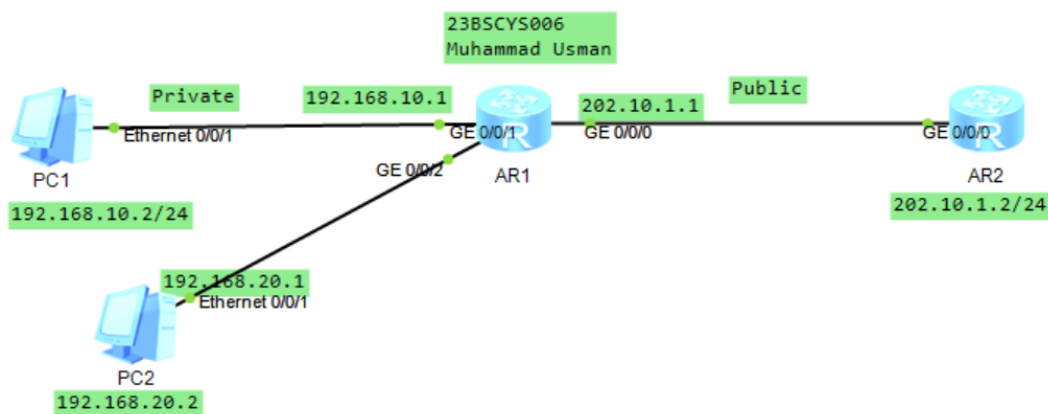
```
Ping 202.10.1.2: 32 data bytes, Press Ctrl_C to break
From 202.10.1.2: bytes=32 seq=1 ttl=254 time=94 ms From
202.10.1.2: bytes=32 seq=2 ttl=254 time=78 ms From
202.10.1.2: bytes=32 seq=3 ttl=254 time=47 ms From
202.10.1.2: bytes=32 seq=4 ttl=254 time=31 ms From
202.10.1.2: bytes=32 seq=5 ttl=254 time=16 ms
```

```
--- 202.10.1.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
0.00% packet loss
 round-trip min/avg/max = 16/53/94 ms
PC>
```

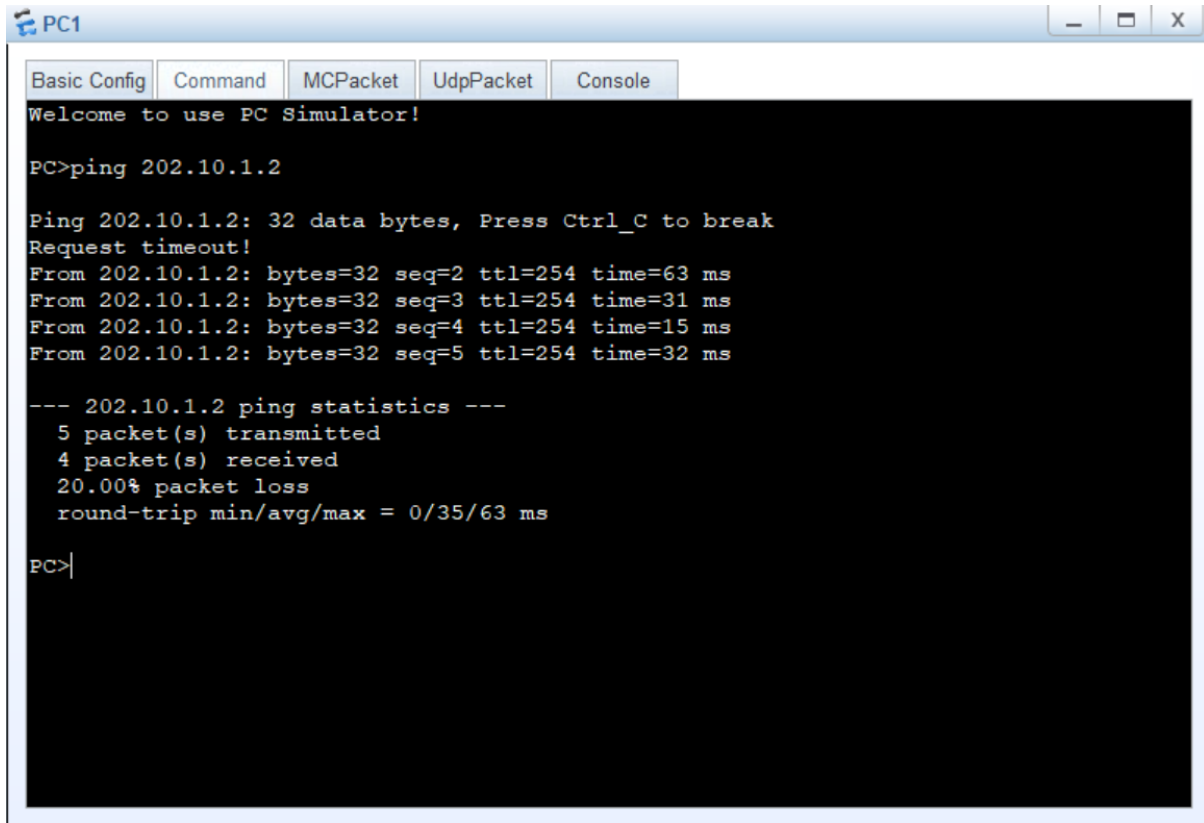
---

## Tasks

Try a different topology and demonstrate your learning.



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The screenshot shows a window titled "PC1" with a menu bar containing "Basic Config", "Command", "MCPacket", "UdpPacket", and "Console". The "Console" tab is active, displaying the following text:

```
Welcome to use PC Simulator!

PC>ping 202.10.1.2

Ping 202.10.1.2: 32 data bytes, Press Ctrl_C to break
Request timeout!
From 202.10.1.2: bytes=32 seq=2 ttl=254 time=63 ms
From 202.10.1.2: bytes=32 seq=3 ttl=254 time=31 ms
From 202.10.1.2: bytes=32 seq=4 ttl=254 time=15 ms
From 202.10.1.2: bytes=32 seq=5 ttl=254 time=32 ms

--- 202.10.1.2 ping statistics ---
 5 packet(s) transmitted
 4 packet(s) received
20.00% packet loss
round-trip min/avg/max = 0/35/63 ms

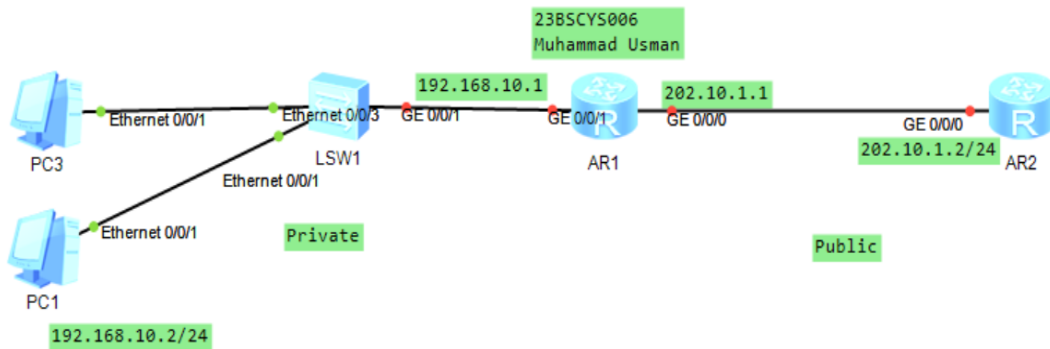
PC>|
```

```
[23BSCYS006-R1]dis nat static
Static Nat Information:
Interface : GigabitEthernet0/0/0
  Global IP/Port : 202.10.1.3/----
  Inside IP/Port : 192.168.10.2/----
  Protocol : ----
  VPN instance-name : ----
  Acl number : ----
  Netmask : 255.255.255.255
  Description : ----

Interface : GigabitEthernet0/0/2
  Global IP/Port : 202.10.1.4/----
  Inside IP/Port : 192.168.20.2/----
  Protocol : ----
  VPN instance-name : ----
  Acl number : ----
  Netmask : 255.255.255.255
  Description : ----

Total : 2
[23BSCYS006-R1]
```

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```
PC3
Basic Config Command MCPacket UdpPacket Console
Welcome to use PC Simulator!
PC>ping 202.10.1.2

Ping 202.10.1.2: 32 data bytes, Press Ctrl_C to break
Request timeout!
From 202.10.1.2: bytes=32 seq=2 ttl=254 time=110 ms
From 202.10.1.2: bytes=32 seq=3 ttl=254 time=46 ms
From 202.10.1.2: bytes=32 seq=4 ttl=254 time=47 ms
From 202.10.1.2: bytes=32 seq=5 ttl=254 time=47 ms

--- 202.10.1.2 ping statistics ---
 5 packet(s) transmitted
 4 packet(s) received
20.00% packet loss
round-trip min/avg/max = 0/62/110 ms
PC>
```

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```
[23BSCYS006-R1]dis nat address-group

NAT Address-Group Information:
-----
Index      Start-address      End-address
-----
1           202.10.1.3          202.10.1.10
-----

Total : 1
[23BSCYS006-R1]
```

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